Homework Week 9

Modern Physics

Gergő Dénes,[1,*] Réka Szilvási,[2,†] Angelo Valli,[1,‡] and Dániel Varjas[1,§]

[1]*Department of Theoretical Physics, Institute of Physics,*
Budapest University of Technology and Economics, Műegyetem rkp. 3., H-1111 Budapest, Hungary
[2]*Department of Nuclear Techniques, Institute of Nuclear Techniques,*
Budapest University of Technology and Economics, Műegyetem rkp. 3., H-1111 Budapest, Hungary

(Dated: April 30, 2026)

For exercises from the textbook, the corresponding number is reported. The text may have been edited for clarity.

Some of the following exercises (possibly with different numbers) may be on the **mini-test** taking place on the Monday of the following week, or on the **midterms**.

ATOMS

Find the electron configuration of the open shell and the values of the total L, S and J using Hund's rules for the following elements:

- C (carbon): [He]$2s^2 2p^2$
- Cl (chlorine): [Ne]$3s^2 3p^5$
- Mn (manganese): [Ar]$3d^5 4s^2$
- Fe (iron): [Ar]$3d^6 4s^2$
- Co (cobalt): [Ar]$3d^7 4s^2$
- Ga (gallium): [Ar]$3d^{10} 4s^2 p^1$
- Sm (samarium): [Xe]$4f^6 6s^2$

SPIN

- **Spin y eigenstates**
  Calculate the eigenstates of the spin-y operator
  $$S_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \frac{\hbar}{2}\sigma_y.$$
- **Spins of general direction**
  Calculate the expectation values of the three spin operators in a state
  $$\psi(\theta,\phi) = \begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} \end{pmatrix}.$$
  The set of these states is called the Bloch sphere.

  Show that it is an eigenstate of the Hamilton operator of a spin in a general direction magnetic field written in polar coordinates $\vec{B} = B(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$:
  $$H = -\vec{B}\cdot\vec{\mu} = \frac{2\mu_B}{\hbar}\vec{B}\cdot\vec{S} = \mu_B B\left(\sin\theta\cos\phi\;\sigma_x + \sin\theta\sin\phi\;\sigma_y + \cos\theta\;\sigma_z\right).$$

* denes.gergo@edu.bme.hu
† szilvasi.reka@ttk.bme.hu
‡ valli.angelo@ttk.bme.hu
§ varjar.daniel@ttk.bme.hu

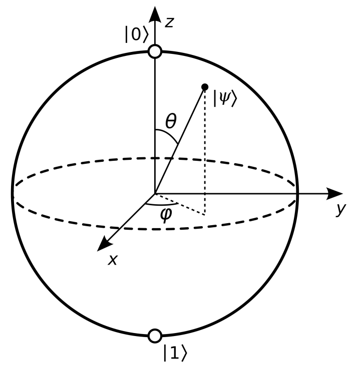

Figure 1. Illustration of the Bloch sphere, here $|0\rangle = (1, 0)$ and $|1\rangle = (0, 1)$.

VARIATIONAL METHODS

- **Binding of the deuteron**

Consider a particle in an attractive potential, described by the Hamiltonian

$$H = -\frac{\hbar^2}{2\mu}\nabla^2 - Ae^{-r/a}.$$

This is a reasonable model for the binding energy of a deuteron (nucleus of ^{2}H) due to the strong nuclear force, taking $\mu = m_\mathrm{p} m_\mathrm{n}/(m_\mathrm{p} + m_\mathrm{n}) = 469.5\,\mathrm{MeV}/c^2$ as the reduced mass of deuteron and parameters $A = 32\,\mathrm{MeV}$ and $a = 2.2\,\mathrm{fm}$. Since the potential is spherically symmetric, most attractive at $r = 0$ and rapidly falls off to zero at large r/a, consider a variational wavefunction which behaves similarly:

$$\psi(r) = Ne^{-\alpha r/2a}$$

characterized by a single *dimensionless* parameter α, and a normalization constant N.

(a) Write the normalization condition and show that it is fulfilled for $N^2 = \alpha^3/(8\pi a^3)$.
Hint: note that the problem is three-dimensional and formulated in polar coordinates (r, θ, ϕ).

(b) Calculate the expression for the ground state energy $E(A, a; \alpha)$ that depends parametrically on α.
Hint: the Laplacian in polar coordinates for a spherically-symmetric function is

$$\nabla^2\psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial\psi}{\partial r}\right]$$

(c) Apply the variational principle and verify that $\alpha_0 = 1.34$ is a solution within three significant digits.

(d) Calculate the ground state energy $E_0 = E(\alpha_0)$.
Hint: as a reference value, the exact solution for this potential yields $E_0 = -2.245\,\mathrm{MeV}$.

- **Variational solution of quartic oscillator first excited state**

Consider the Hamiltonian of a particle in a 1D quartic potential:

$$H = K + V = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \lambda x^4.$$

Find the minimal energy wavefunction using a harmonic oscillator trial wavefunction

$$\psi_1(x) = \left(\frac{\alpha}{\pi}\right)^{1/4}\sqrt{2\alpha}\,x\,e^{-\frac{1}{2}\alpha x^2},$$

where α is a free parameter. Calculate the value of α that minimizes the energy expectation value, and find the minimal energy.

- **Stark effect of the hydrogen L shell**
The Stark effect is the shift of atomic spectral lines in the presence of an electric field, it is analogous to the Zeeman effect which happens when a magnetic field is applied. Here we consider a hydrogen atom in an electric field in the z direction described by the Hamiltonian

$$H = -\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{4\pi\epsilon_0 r} + \mathcal{E}z = H_0 + V',$$

where H_0 is the hydrogen Hamiltonian without electric field and $V' = \mathcal{E}z$.

Our goal is to calculate the Stark shift of the L-shell ($n = 2$) states ψ_{200} and ψ_{210}. We use a trial wavefunction

$$|\psi\rangle = \alpha\,|\psi_{200}\rangle + \beta\,|\psi_{210}\rangle\,, \quad \psi_{200}(r) = \frac{1}{4\sqrt{2\pi a_0^3}}\left(2 - \frac{r}{a_0}\right)e^{-\frac{r}{2a_0}}, \quad \psi_{210}(r,\theta) = \frac{1}{4\sqrt{2\pi a_0^3}}\frac{r}{a_0}e^{-\frac{r}{2a_0}}\cos\theta.$$

(a) What is the expression for the expectation value $\langle H_0\rangle_\psi$? (Remember that the energy of hydrogen eigenstates only depend on n.)

(b) Calculate the expression for the electric potential energy expectation value $\langle V'\rangle_\psi$ that depends parametrically on α and β.
Write everything in spherical polar coordinates and use the integral identities $\int_0^\infty x^n e^{-x/a}dx = n!a^{n+1}$, $\int_0^\pi \sin x \cos^2 x\, dx = 2/3$. *Only one of the matrix elements is nonzero, you can use symmetry arguments to show that the rest must vanish.*

(c) Calculate the expression for the energy expectation value $E(\alpha,\beta) = \langle H\rangle_\psi$, write the energy in the matrix form as $E = v^\dagger H v$ where $v = (\alpha,\beta)^T$ is the column vector formed of the unknown coefficients, and H is a 2×2 matrix. Find its eigenvalues as a function of $\mathcal{E}$, these give the energies of the two variational states. You should find that the two energies move linearly as a function of $\mathcal{E}$ in opposite direction.

(d) Include the other two $2p$ states in the calculation and use a trial wavefunction as a linear combination of all four L-shell states. You should find that all other matrix elements vanish, the result is essentially the same, with no change in the energies of the other two states.

SPECIAL RELATIVITY

- **Velocity addition: $p + \bar{p}$ collision / Ex. 2.32**
A proton and an antiproton are moving toward each other in a head-on collision. If each has a speed of $u = 0.8\,c$ with respect to the collision point, how fast are they moving with respect to each other?

- **Increase of momentum with speed / Ex. 2.60**
A particle initially has a speed of $u_0 = 0.5\,c$ and momentum p_0. At what speed u does its momentum p increase by (a) 1 %, (b) 10 %, (c) 100 %?

- **Length contraction / Ex 2.24**
A spaceship of length $\ell_0 = 40\,\mathrm{m}$ when measured in a reference frame co-moving with the ship, is observed to be $\ell = 20\,\mathrm{m}$ long when measured in a reference frame in which it is in motion (at a constant velocity). How fast is it moving?

- **Velocity addition: shooting a proton gun / Ex. 2.31**
A spaceship is moving at a speed of $v = 0.84\,c$ away from an observer at rest. A boy in the spaceship shoots a proton gun with protons having a speed of $u' = 0.62\,c$. What is the speed of the protons measured by the observer at rest when the gun is shot (a) away from the observer and (b) toward the observer?

- **Relative Motion / Ex. 2.4**
A swimmer wants to swim straight across a river with current flowing at a speed of $v_1 = 0.350\,\mathrm{m/s}$. If the swimmer swims in still water with speed $v_2 = 1.25\,\mathrm{m/s}$, at what angle should the swimmer point upstream from the shore, and at what speed will the swimmer swim across the river?

- **Spacetime / Ex. 2.13**
Two events occur in an inertial system K as follows:

1. $x_1 = a$, $t_1 = 2a/c$, $y_1 = 0$, $z_1 = 0$

2. $x_2 = 2a$, $t_2 = 3a/2c$, $y_2 = 0$, $z_2 = 0$

for some constant a. In what frame K' will these events appear to occur at the same time? Describe the motion of system K'.

- **Length contraction and time dilation / Ex 2.19**
  A rocket ship carrying passengers blasts off to go from New York to Los Angeles, a distance of about $d = 5000\,\mathrm{km}$ (as measured in the reference frame at rest on Earth). (a) How fast must the rocket ship go to have its own length ℓ_0 (in the reference frame co-moving with it) shortened by $1\,\%$ to ℓ' (as measured in Earth's reference frame)? (b) Ignore effects of general relativity and determine how much time the rocket ship's clock and the ground-based clocks differ when the rocket ship arrives in Los Angeles.

- **Twin Paradox / Ex. 2.20**
  Astronomers discover a planet orbiting around a star similar to our sun that is $\ell = 20\,\mathrm{ly}$ away (as measured in a reference frame at rest on Earth). How fast must a rocket ship go if the round trip is to take no longer than $t' = 40\,\mathrm{y}$ in time for the astronauts aboard? How much time (t) will the trip take as measured on Earth?

- **A more challenging Lorentz transformation: shooting a proton gun / Ex. 2.30**
  Two systems K and K' synchronize their clocks at $t = t' = 0$ when their origins are aligned as system K' passes by system K along the x axis at relative speed $v = 0.8\,c$. At time $t = 3\,\mathrm{ns}$, Frank (in system K) shoots a proton gun having proton speeds of $u_x = 0.98\,c$ along his x axis. The protons leave the gun at $x_1 = 1\,\mathrm{m}$ and arrive at a target 120 m away, at $x_2 = 121\,\mathrm{m}$. Determine the event coordinates (x_1, t_1) and (x'_1, t'_1) of the gun firing, and the event coordinates (x_2, t_2) and (x'_2, t'_2) of the protons arriving, as measured by observers in both systems K and K', respectively.

- **Velocity addition: relative velocity of galaxies / Ex. 2.35**
  Three galaxies (A)ndromeda, (B)utterfly, and (C)ocoon are aligned along an axis in the order A, B, C. An observer in galaxy B is in the middle and observes that galaxies A and C are moving in opposite directions away from him, both with speeds $u = u_A = u_C = 0.60\,c$ (whereas $u_B = 0$). What is the speed U'_B and u'_C of galaxies B and C as observed by someone in galaxy A?